

Vaibhav Bhosale

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EDUCATION

- 2021–2026 **Ph.D.**, *Computer Science*, Georgia Institute of Technology, Atlanta, United States.
Advisors: Dr. Ada Gavrilovska, Dr. Ketan Bhardwaj
- 2019–2021 **M.S.**, *Computer Science*, Georgia Institute of Technology, Atlanta, United States.
- 2013–2017 **B.Tech. with Honors**, *Computer Science & Engineering*, Indian Institute of Technology Bombay, Mumbai, India.

PUBLICATIONS

- **The Storage Wall in LEO Compute Cloud.** Vaibhav Bhosale, Ada Gavrilovska, Ketan Bhardwaj. In 18th ACM Workshop on Hot Topics in Storage and File Systems 2026 (HotStorage 2026), 2026 [Collocated with SOSP 2026].
- **Toward Trainable AI Agents for Satellite Operations**. Haley Calderon, Vaibhav Bhosale, Ketan Bhardwaj, Ada Gavrilovska. Small Satellite Conference (SmallSat 2026), 2026.
- **Lighting Up the Moon: A Scheduling Framework for Cislunar Space Based Solar Power Orchestration.** Vaibhav Bhosale, Andrew Woo, Ada Gavrilovska, Christopher R. Valenta, Ketan Bhardwaj. Small Satellite Conference (SmallSat 2026), 2026.
- **Keeping Up with the Trends: Identifying Shifts in LEO Satellite Usage Patterns.** Stepan Kravtsov, Sherif Abdelrazek, Vaibhav Bhosale, Ahmed Saeed. In 4th ACM Workshop on LEO Networking and Communication (LEO-NET 2026), 2026 [Collocated with Sigcomm 2026].
- **Assessing LEO Satellite Networks for National Emergency Failover.** Vaibhav Bhosale, Ying Zhang, Sameer Kapoor, Robin Kim, Miguel Schlicht, Muskaan Gupta, Ekaterina Tumanova, Zachary Bischof, Fabián E. Bustamante, Alberto Dainotti, Ahmed Saeed. In Proceedings of the 2025 ACM on Internet Measurement Conference (IMC 2025), 2025.
- **Ushering in the Low Earth Orbit Compute Cloud Era.** Ketan Bhardwaj, Vaibhav Bhosale, Ada Gavrilovska. In Computer Magazine, 2025.
- **Are LEO Networks the Future of National Emergency Failover? – A Quantitative Study and Policy Blueprint.** Vaibhav Bhosale, Ying Zhang, Sameer Kapoor, Robin Kim, Miguel Schlicht, Muskaan Gupta, Ekaterina Tumanova, Zachary Bischof, Fabián E. Bustamante, Alberto Dainotti, Ahmed Saeed. In The Research Conference on Communications, Information and Internet Policy (TPRC53), 2025.
- **Krios: Scheduling Abstractions and Mechanisms for Enabling a LEO Compute Cloud.** Vaibhav Bhosale, Ada Gavrilovska, Ketan Bhardwaj. In Proceedings of the 2024 ACM Symposium on Cloud Computing (SoCC 2024), 2024.
Awarded Best Paper
- **A Characterization of Route Variability in LEO Satellite Networks.** Vaibhav Bhosale, Ahmed Saeed, Ketan Bhardwaj, Ada Gavrilovska. In Passive and Active Measurement Conference (PAM 2023), 2023.
Invited presentation at IETF/IRTF Measurement and Analysis for Protocols Research Group (MAPRG)
- **PaddleSats: Attitude Control and Station-Keeping for Ultra-Low Density SSP Satellites.** Vaibhav Bhosale, Jonathan Dolan, Grishma Kalepu, Deeksha Manjunath, Gregory Durgin. In IEEE International Conference on Wireless for Space and Extreme Environments (WiSEE 2023), 2023.
Awarded Best Paper
- **Don't Let Your LEO Edge Fade at Night.** Vaibhav Bhosale, Ketan Bhardwaj, Ada Gavrilovska. In 1st Workshop on Hot Topics in Systems Infrastructure (HotInfra 2023), 2023 [Collocated with ISCA 2023].
- **Astrolabe: Modeling RTT Variability in LEO Networks.** Vaibhav Bhosale, Ketan Bhardwaj, Ahmed Saeed. In 1st ACM Workshop on LEO Networking and Communication (LEO-NET 2023), 2023 [Collocated with Mobicom 2023].
- **Efficient Cross-Frequency Beam Prediction in 6G Wireless Using Time Series Data.** Vaibhav Bhosale, Navrati Saxena, Ketan Bhardwaj, Ada Gavrilovska, Abhishek Roy. In IEEE 2024 International Symposium on Networks, Computers and Communications (ISNCC'24).
- **Toward Loosely Coupled Orchestration for the LEO Satellite Edge.** Vaibhav Bhosale, Ketan Bhardwaj, Ada Gavrilovska. In 3rd USENIX Workshop on Hot Topics in Edge Computing (HotEdge 2020), 2020.

PATENTS

- **Techniques for Overriding Libraries for Workloads in a Cloud-Computing Environment.** US Patent 11838354 (2023)
- **Spatial and Frequency Domain Beam Management Using Time Series Information.** US Patent 18369488 (2024)

SERVICE

- **Journal Reviews:** 2026 IEEE TNSM (2), 2026 IEEE TMC, 2024 IEEE TPDS, 2023 IEEE/ACM ToN, 2022 IEEE JSAC.
- **Conference Reviews:** IEEE ICC 2026, IEEE ICC 2024.
- **Workshop Reviews:** ACM LEO-NET 2026.
- **Artifact Evaluation Committee:** SOSP 2025, Eurosys 2025, SOSP 2024, OSDI 2023, ATC 2023
- **Shadow PC:** Eurosys 2024

AWARDS AND ACADEMIC ACHIEVEMENTS

- Runner-up at the Annual College of Computing Poster Symposium, Georgia Tech. (2025)
- Finalist at Qualcomm Innovation Fellowship (45 out of ~200). (2023)
- Finalist (top 5) at Microsoft One Week Hackathon in the Bengaluru Science Fair for a Speech-based Drawing Assistant (2018)
- Runners up at Mimamsa, the largest collegiate science quiz in India conducted by IISER Pune. (2016)
- Winner at Bazinga Mathematics, the largest Mathematics quiz at IIT Bombay (80 teams participated). (2016)
- Secured **All India Rank 161** in JEE-Advanced out of 1.5 million students. (2013)
- Amongst **top 1%** in the National Standard Examinations in Physics and Astronomy. (2013)
- Awarded **National Talent Search Exam** scholarship conducted by Government of India (selection ratio $\approx 0.1\%$). (2009)

PROFESSIONAL EXPERIENCE

- Summer 2024 **Research Intern**, *Microsoft Research*, Future of Scalable Software Engineering (FOSSE), Mentors: Dr. Peter Bodik, Iker Llorens.
- Summer 2022 **5G Research Intern**, *Mediatek Inc.*, Advanced Communication Technology, Mentor: Dr. Abhishek Roy.
- Summer 2021 **Research Intern**, *Microsoft Inc.*, Azure for Operators (AfO), Mentor: Dr. Sanjeev Mehrotra.
- Summer 2020 **Senior Backend Developer Intern**, *IBM Inc.*, Cloud and Cognitive Services.
- 2017–2019 **Software Engineer**, *Microsoft India Development Center*, Bengaluru, Outlook Rest API.
- Summer 2016 **Software Development Intern**, *Microsoft India Development Center*, Bengaluru.

TEACHING EXPERIENCE

- **Head Teaching Assistant:** Artificial Intelligence (Fall 2020, Spring 2021)
Led a team of 21 Graduate Teaching Assistants, overseeing the planning and execution of course structure, assignments, examinations, and enhancing the overall student experience.
- **Graduate Teaching Assistant:** Distributed Computing (Fall 2023, Fall 2021), Artificial Intelligence (Fall 2019, Spring 2020)
- **Undergraduate Teaching Assistant:** Artificial Intelligence (Spring 2017), Computer Architecture (Autumn 2016), Computer Programming & Utilization (Summer 2017)

LEADERSHIP ROLES

- 2023-24 **School of Computer Science Graduate Student Association**, *President*.
Led an executive team comprising of 8 grad students overseeing academic, welfare, administration, and social activities for the graduate student body, while managing a budget exceeding \$50,000
- 2022-23 **School of Computer Science Graduate Student Association**, *Finance Chair*.
- 2016-17 **Students' Technical Activities Body**, *Undergraduate Events Nominee*.
Led a 3-tier team consisting of 20 managers and 100+ volunteers to handle 200+ events with a INR 1.5M budget for 7000+ students.
- 2015-16 **Maths & Physics Club**, *Manager*.